



Muhammad Asad

Petroleum Engineer | Researcher | Data & ML Enthusiast

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About Me

Certified Cloud Native Developer (PIAIC) with a Petroleum Engineering background and a strong transition into cloud computing, data analytics, and applied machine learning. Experienced in research-driven problem solving, workflow automation, and cross-domain innovation. Skilled in Python, Docker, and web development, integrating engineering knowledge with modern IT solutions to improve efficiency and data-driven decision-making.

Education

Certified Cloud Native Developer

Presidential Initiative for AI & Computing (PIAIC)

Batch 36 | 2024

Bachelor in Petroleum Engineering

NED University of Engineering and Technology

Graduated, 2023

CGPA: 3.619/4

Intermediate in Pre-Engineering

Adamjee Govt. Science College

Karachi Board

Passed, 2019

Marks 77.00%

A Grade

Matriculation in Science

A A Tutor Academy

Karachi Board

Passed, 2017

Marks 88.47%

A-One

Work Experience

February 2024 – Present

Research Analyst.

Cloud Innovators Solution (CIS), Park Avenue, Shahra-e-Faisal, Karachi



- Apply data analytics and ML pipelines for operational optimization.
- Develop project plans and documentation using MS Project; streamline workflows through structured planning.
- Conduct advanced statistical analysis and create insightful visualisations to support research findings.
- Implement machine learning models to extract insights and improve operational performance across projects.
- Prepare technical documentation, research reports, and dashboards for multi-disciplinary teams

May 2021 – August 2023

Research Assistant.

NED University of Engineering and Technology, University Road, Karachi



- Conducted literature reviews, experimental testing, and data analysis on shale swelling, drilling fluid rheology, and asphaltene precipitation.
- Implemented ML algorithms to predict fluid properties and stability parameters.
- Co-authored 5+ peer-reviewed papers (Chemical Papers, Arabian Journal for Science & Engineering).

October 2022 – November 2022

Engineering Intern.

Oil & Gas Development Company (OGDCL), Tando Jam, Hyderabad



- Visited well sites to observe jet pump operations, understand surface and subsurface equipment, and assist in reverse-out procedures.

June 2022 - July 2022

Data Analyst Intern.

GlobalShala, Sponsored by Saint Louis University, Virtually



- Proficient in audience-tailored data presentations, meaningful content analysis, basic data analytics tools, and utilising visual perception for effective communication.

Academic Projects

July 2022 – August 2023

ML-Driven Prediction of Reservoir Fluid Properties (Halliburton FYDP)

- Machine Learning-Driven Prediction of Reservoir Fluid Property for Enhanced Reservoir Management.

January 2023 – July 2023

Cloud Microservices Deployment (PIAIC)

- Developed and deployed a containerized microservice on Docker + Kubernetes using Flask API.

May 2021 – September 2021

Web Dashboard for Research Visualization

- Built an interactive data visualization dashboard (HTML, CSS, JavaScript, Plotly)

Publication

Publications & Conference Contributions:

Co-authored five+ peer-reviewed publications in *Chemical Papers* and *Arabian Journal for Science and Engineering*, covering shale stability, green drilling fluids, and AI-driven reservoir modeling.

- **SPE ADIPEC 2023 Conference Contribution:** Taro Mucilage as Green Inhibitor in Water-based Drilling Fluid
- **ACS Acknowledged Work:** Carbon Capture and Storage (CCS) in Oil and Gas
- **Geothermal Rising Student Contest (Abstract Selected):** Enhanced Geothermal Systems: Global Challenges & Strategies

A full list is available on request or via Google Scholar.

Extra-Curricular Activities

International Switch Energy Case

2020 – 2022

- Certificate of participation in the 2020 International Switch Energy Case Competition, qualifying for the semi-finals in 2022.

OPENAI WHISPER, GPT3 Hackathon

December 2022

Lablab AI

- Certificate for outstanding performance, attendance, and completing the hackathon.

Technical Domain Training

September 2021 – May 2022

National Freelance Training Program (NFTP)

- Completion Certificate of training in Technical Domain (Web Development and freelancing)

Extreme Commerce by Sunny Ali,

October 2020

- Certificate of completion of Virtual Assistance (VA) and FFL of Amazon from Extreme Commerce, presented by Sunny Ali.

Coursera

October 2021 – October 2025

- Certificate in Microsoft Excel (Essential, Intermediate I, Intermediate II)
- Certificate in Google Data Analytics
- Certificate in Google IT Automation with Python
- Certificate in Google Advance Data Analytics



Achievements

- Co-authored 5 peer-reviewed papers (topics: shale stability, green drilling fluids, AI in reservoir modeling).
- Selected as University Representative at the SPE ADIPEC University Programme, and we secured 2nd place out of 13 international university teams.
- Awarded by NED Alumni Association of Houston Scholarship.
- Awarded by the Chief Minister of Sindh, Syed Murad Ali Shah, for securing an A-One grade.



Memberships

- | | |
|---|------------------------------------|
| • American Chemical Society (ACS) | Community Associate 2023 - Present |
| • Society of Petroleum Engineering (SPE) | Executive Member 2019 - Present |
| • Student Energy NED Chapter (SENC) | Executive Member 2021 - 2022 |



Skills

Petroleum Software

- MBAL
- ImageJ
- Fiji

Front-End Web Developer

- HTML5
- CSS3 + Bootstraps
- JavaScript
- Python
- R

MS Office

- Microsoft Word
- Microsoft PowerPoint
- Microsoft Excel
- Microsoft Project



Languages

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|-----------|--------|-------------------|
| • English | • Urdu | • Chinese (Basic) |
| | | • Italian (Basic) |